

See also

[Abdominal pain - acute](#)
[Acute pain management](#)

Key points

- The diagnosis of intussusception requires a high index of suspicion. Consider intussusception in infants and children with intermittent distress, pallor, vomiting or isolated unexplained lethargy
- Fluid resuscitation is important prior to sedation and reduction
- Delayed presentation of intussusception can manifest as small bowel obstruction, bowel perforation, peritonitis and/or shock
- Ultrasound is the initial study of choice
- Ileocolic intussusception necessitates rapid diagnosis and a collaborative treatment approach involving emergency medicine, radiology and surgery

Background

Intussusception is the invagination (telescoping) of a proximal segment of bowel into the distal bowel lumen. The most common site is a segment of ileum moving into the colon through the ileo-caecal valve. This process leads to bowel obstruction, venous congestion and eventually bowel wall ischaemia. Perforation can occur and lead to peritonitis and shock

- May occur at any age, but most commonly between 3 months and 2 years of age, with peak incidence at 4-9 months of age
- Most cases are idiopathic, with one-third of the patients having a recent viral illness
- In younger infants, the clinical presentation may be atypical, often more acute and rapidly progressive
- In older children, a pathological lead point may be the cause

Assessment

History

- Children most commonly present to the ED with paroxysmal (colicky) abdominal pain and/or vomiting (70% of presentations)
 - The typical paroxysmal pain is described as severe, cyclic or cramping. The pain episodes usually last 4-5 minutes
 - Episodes can recur within minutes to hours and may increase in frequency over the next 12-24 hours
 - Vomiting is usually a prominent feature, **bile-stained vomiting** is a late sign and indicates a bowel obstruction
- Intermittent pain or distress, the child may appear very well between episodes
- Poor feeding
- Pallor, especially during episodes
- Lethargy may be the only presenting symptom. It may be profound, episodic or persistent
- Diarrhoea may occur initially and can lead to a misdiagnosis of gastroenteritis. **Rectal bleeding** or the classic **"red currant jelly"** stool are late, less common signs suggesting bowel ischemia and infarction
- The triad of intermittent abdominal pain, palpable abdominal mass and red currant jelly stools occurs in less than 15% of children

Additional risk factors

- More frequent in males than females (3:1 ratio)
- Recent intussusception (may present with more subtle symptoms)
- [Henoch Schönlein Purpura](#)
- Rare associations: Meckel's diverticulum, lymphoma, luminal polyps eg Peutz Jegher Syndrome, recent bowel surgery
- Current rotavirus vaccines are not linked to intussusception

Examination

- Abdominal distension** suggests bowel obstruction
- Tenderness or guarding** may suggest perforation and peritonitis
- Abdominal mass may be felt, typically a sausage shaped mass in the right abdomen, crossing the midline in the epigastrium or behind umbilicus
 - may be subtle and examination is best performed when the child is settled in between episodes
- Inspect the nappy and perianal region. A rectal examination is rarely indicated
- Infants may present with **hypovolaemic shock**

Management

- Resuscitation and analgesia should precede investigation.** See flowchart below
- Secure IV access for all children suspected to have intussusception before diagnostic imaging
- Treat [hypovolaemic shock](#) with IV boluses of 0.9% saline
- Start IV fluids to optimise circulatory volume
- Ensure adequate analgesia, usually intranasal fentanyl or IV morphine, see [Acute pain management](#)
- Keep nil orally. Pass nasogastric tube if bowel obstruction or perforation on AXR, or if planning transfer by air

Investigations

Ultrasound scan

- High sensitivity (>98%) and specificity (>98%) when performed by an experienced paediatric ultrasonographer
- Point of Care Ultrasound (POCUS) can be used to confirm the diagnosis of intussusception by appropriately trained clinicians (95% sensitivity and 98% specificity). It should not be used to exclude the diagnosis
- If no bowel abnormality is seen, extend the examination to full abdomen to rule out other causes for pain
- Ileo-ileal intussusception, involving invagination of two small bowel segments, may present with mild symptoms. This type of intussusception is more likely to resolve spontaneously and rarely requires enema reduction, as ischaemia is rare. This is generally the manifestation of viral illness and usually self resolves if the affected area is small (2-3cm)

Abdominal X-Ray

- Perform AXR **only** if there are signs of obstruction or perforation
- AXR may be used in centres without ready access to ultrasound, although this should not delay transfer to appropriate services.
- A normal AXR does not exclude intussusception (sensitivity <50%)
- Signs suggesting intussusception on an abdominal x-ray include
 - an abnormal gas pattern, with an empty right lower quadrant and visible soft tissue mass in the upper abdomen
 - a soft tissue mass surrounded by a crescent lucency of bowel gas (crescent sign)
 - lack of faecal material in the large bowel
 - signs of **small bowel obstruction**
 - pneumoperitoneum** indicating bowel perforation

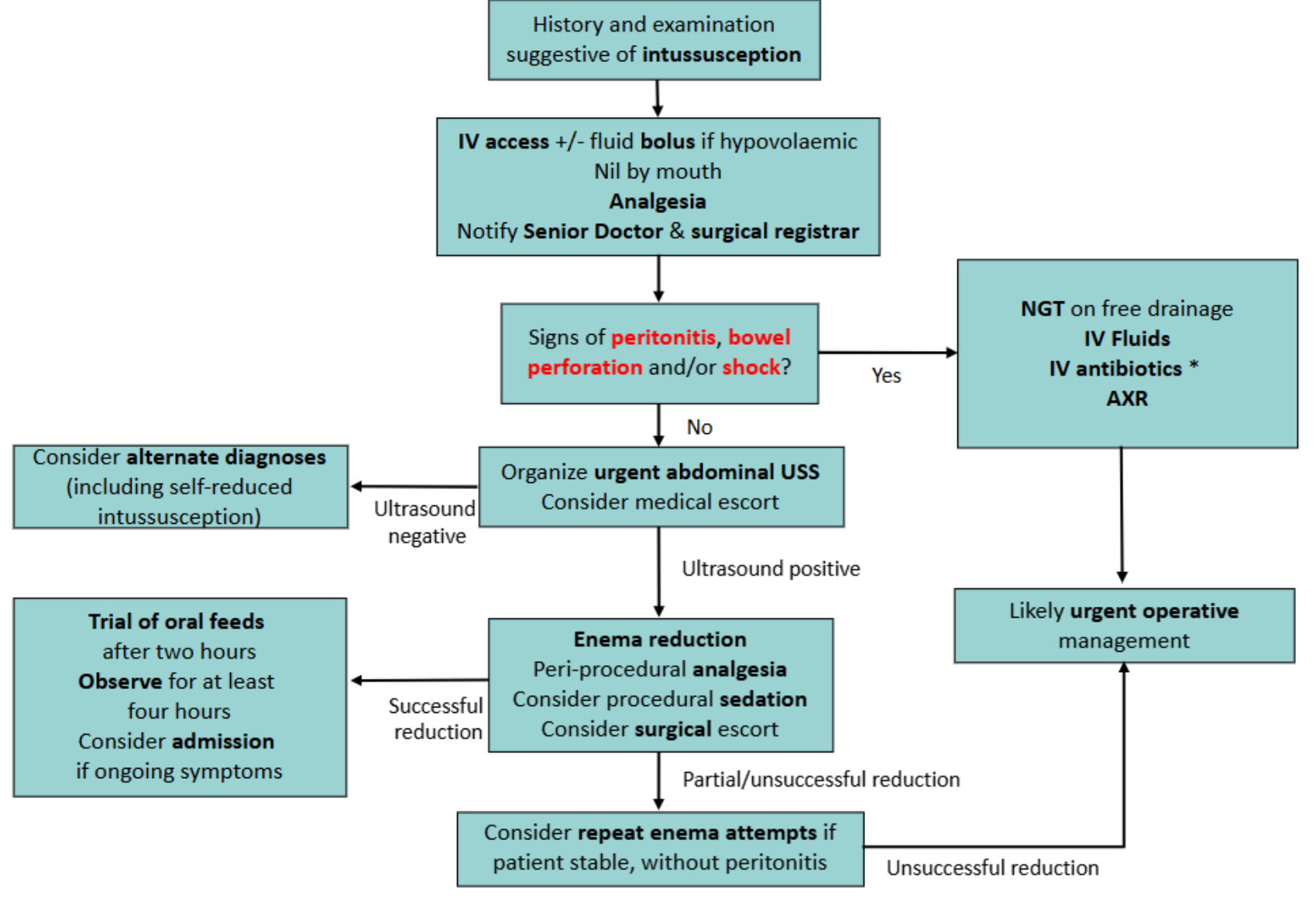
Blood tests

- Blood glucose
- Venous gas, FBE and UEC if the child looks unwell
- Blood group and hold prior to theatre

Contrast/gas enema

- The enema may be used diagnostically and therapeutically in consultation with a surgical team.
- There is a small risk of **bowel perforation** and **bacteraemia** during the gas enema. Therefore, the enema is performed where paediatric surgery is available in case of the need for laparotomy. A surgical doctor, as well as a suitably trained nurse, may accompany the child with appropriate resuscitation equipment
- Appropriate peri-procedural analgesia is necessary, is usually intranasal fentanyl or IV morphine, see [Acute pain management](#)
- Procedural sedation by an experienced clinician should be strongly considered, as this may improve the success rate of non-operative reduction, both hydrostatic and pneumatic enema
- Routine prophylactic antibiotics do not decrease post-reduction complications and are therefore unnecessary
- Success rate is more than 80%
- Recurrences most commonly occur within the first 24-48 hours
- Repeated enemas may be considered and may increase success rates by 10% with few complications
- Contraindicated** in **peritonitis**, **shock**, radiological evidence of **bowel perforation**, or an unstable clinical condition

Approach to management



* Antimicrobial recommendations may vary according to local antimicrobial susceptibility patterns. Refer to [local guidelines](#)

Follow up after non-operative reduction

- In most centres, admission under the surgical team post reduction is routine
- In discussion with the surgical team, it may be safe to trial a feed 2 hours after reduction and consider discharge from ED a further 2 hours after a successful feed

Consider consultation with local paediatric team when

All children with suspected intussusception should be discussed with a paediatric surgical team

Consider transfer when

Child requiring care beyond the capability of the hospital

Note: when transferring infants or children with possible surgical conditions, ensure they have adequate analgesia, venous access and intravenous fluids prior to transfer, as third space losses can be large and lead to haemodynamic collapse. Insert a nasogastric tube on free drainage if transferring by air.

For emergency advice and paediatric or neonatal ICU transfers, see [Retrieval Services](#)

Parent information

[Abdominal pain](#)
[Reducing your child's pain during investigations and procedures](#)
[Pain relief for children](#)
[Intussusception Parent Information \(WA\)](#)

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